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Derivatives for Special Functions

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Deriving the Addition Rule

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- $= k \cdot \frac{du}{dx}$

Example of Addition and Constant Rules

Consider the functions of $j(x) = 2x^2 + 4x$, $f(x) = x^2$, and $g(x) = x$.

We will show that $\frac{dj(x)}{dx} = 2\frac{df(x)}{dx} + 4\frac{dg(x)}{dx}$

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Deriving the Chain Rule

Let f and g be differentiable functions.

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- $= \frac{dg(u)}{du} \frac{du}{dx}$, where $u = f(x)$

Example of the Chain Rule

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- Break this function into two functions: $g(x) = x^2$ and $f(x) = 3x + 6$.
- Apply the chain rule: $\frac{dg(u)}{du} \frac{df(x)}{dx} = 2(3x + 6)3 = 6(3x + 6)$

Partial Statement of the Inverse Function Theorem

We can use the chain rule to compute the derivative of the inverse of a function. Recall that for an inverse of a function

$f(f^{-1}(x)) = x$. Using the chain rule we get:

$$\blacksquare \quad \frac{df(f^{-1}(x))}{dx} = \frac{df(u)}{du} \frac{df^{-1}(x)}{dx}, \text{ where } u = f^{-1}(x)$$

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- $\frac{df^{-1}(x)}{dx} = \frac{1}{f'(f^{-1}(x))}$
- Consider $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$, $f(x) = x^2$ and $f^{-1}(x)$, then $f'(x) = 2x$ and $\frac{df^{-1}(x)}{dx} = \frac{1}{f'(f^{-1}(x))} = \frac{1}{2x^{1/2}}$

Deriving Product Rule

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- $= (\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h})(\lim_{h \rightarrow 0} g(x+h)) +$
 $(\lim_{h \rightarrow 0} f(x))(\lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h})$ (Pulling out Common Terms)

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 $(\lim_{h \rightarrow 0} f(x))(\lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h})$ (Pulling out Common Terms)
- $= f'g + fg'$

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- Apply Product and Chain Rule:
 $\frac{df(x)}{dx} = (x + 2)^{-2} - 2(x + 1)(x + 2)^{-3}$

Deriving the Power Rule

$$\blacksquare \quad \frac{dx^n}{dx} = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} = \lim_{h \rightarrow 0} \frac{x^n + nhx^{n-1} - x^n}{h} = nx^{n-1}$$

Applying the Power Rule

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- $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = \lim_{h \rightarrow 0} \frac{-1 + 1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \dots}{h} = 1$

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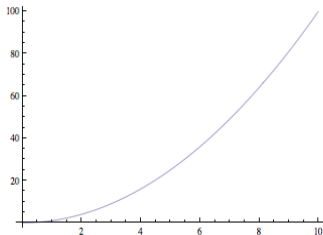
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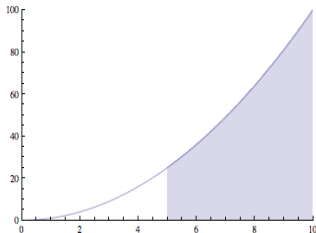
Definite Integrals

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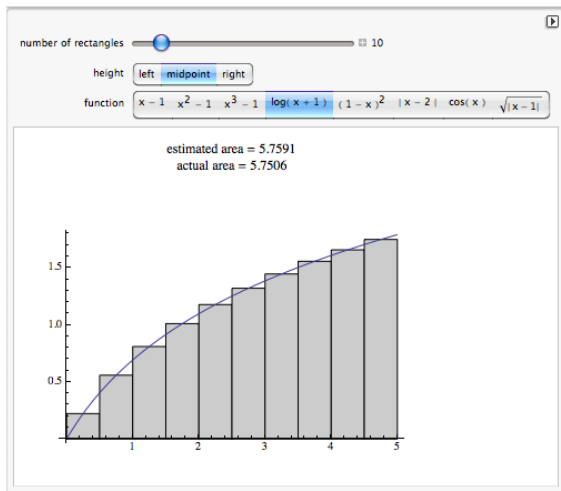


Definite Integrals

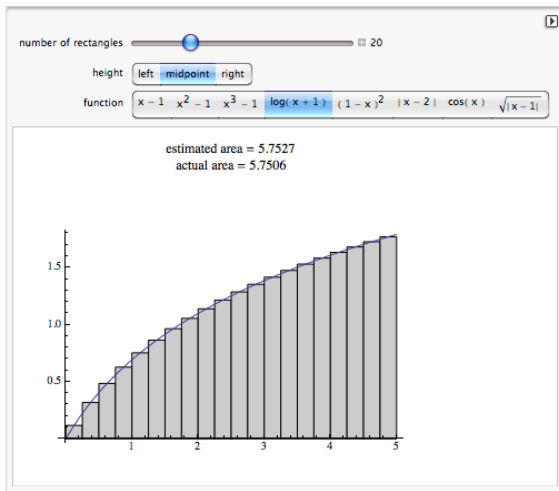
Consider the function $f(x) = x^2$. A definite integral, calculated, for example, between 5 and 10, allows us to calculate the area under the graph between points $x = 5$ and $x = 10$. This is expressed as $\int_5^{10} f(x) dx$.



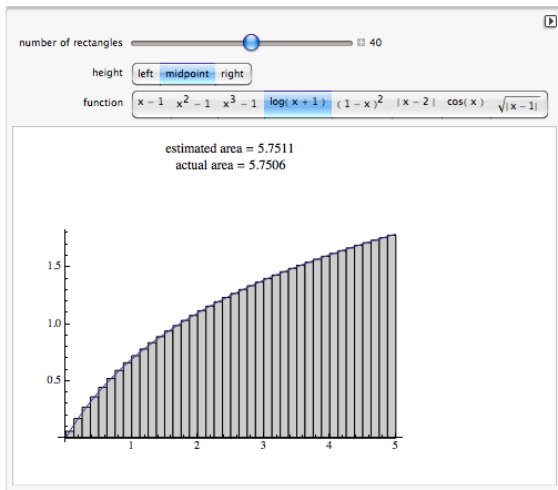
Riemann Sums



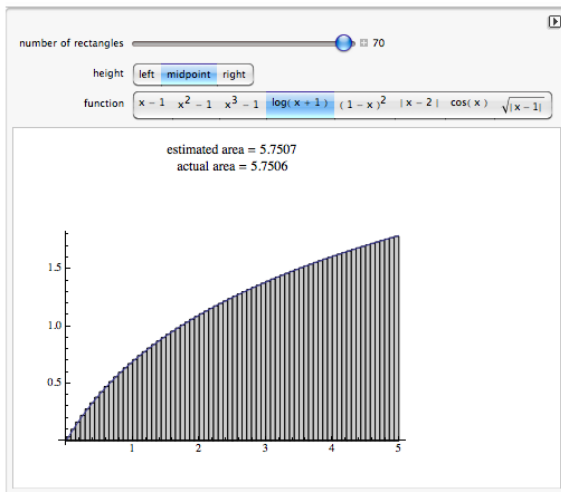
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Indefinite Integrals

If $f(x)$ is given then any function $F(x)$ such that $F'(x) = f(x)$ is called an indefinite integral of $f(x)$, or the anti-derivative. Note that there are infinitely many anti-derivatives of a function $f(x)$ since they can differ by a constant. We denote anti-derivatives by $\int f(x)dx$. Rules for finding anti-derivatives:

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$
- $\int e^x dx = e^x + C$
- $\int a^x \ln(a) dx = a^x + C$
- $\int f'(x) e^{f(x)} dx = e^{f(x)} + C$
- $\int \frac{f'(x)}{f(x)} dx = \ln[f(x)] + C$

The Fundamental Theorem of Calculus

If $f(x)$ is continuous on $[a, b]$, and $F(x)$ is the anti-derivative of $f(x)$, then $\int_a^b f(x)dx = F(b) - F(a)$. The integral of f from a to b is nothing else than the anti-derivative, F , computed at b minus the anti-derivative, F , at a .

The Integral as a Linear Operator

$$\int [af(t) + bg(t)] dt = a \int f(t) dt + b \int g(t) dt$$

- Define the anti-derivatives $ay(x) = a \int \frac{dy}{dx} dx$ and $bz(x) = b \int \frac{dz}{dx} dx$

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- These statements imply that

$$\int (a \frac{dy}{dx} + b \frac{dz}{dx}) dx = a \int \frac{dy}{dx} dx + b \int \frac{dz}{dx} dx$$

Properties of Integrals

- $\int_a^b [f(t) + g(t)] dt = \int_a^b f(t) dt + \int_a^b g(t) dt$
- $\int_a^b \lambda f(t) dt = \lambda \int_a^b f(t) dt$
- $\int_a^c f(t) dt = \int_a^b f(t) dt + \int_b^c f(t) dt$
- $\int_a^b f(t) dt = - \int_b^a f(t) dt$
- $\int_a^a f(t) dt = 0$

Integration by Substitution

To find an integral of the form $\int_a^b f(g(x))g'(x)dx$ one can apply the substitution rule:

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$$

where $u = g(x)$ and $du = g'(x)dx$.

Integration by Substitution: Example

Find $\int (x+1)^{10} dx$.

For this example $f(x) = x^{10}$, $g(x) = x+1$, and $g'(x) = 1$. Let $u = x+1$. This implies $du = dx$.

Substituting this into the equation above we know have the integral $\int u^{10} du = \frac{u^{11}}{11}$.

Substituting this back in for u we find that $\int (x+1)^{10} dx = \frac{(x+1)^{11}}{11}$. This is easier than expanding the function and finding the integral.

Example 2

Consider the integral $\int (\frac{1}{2\pi} x e^{\frac{x^2}{2}}) dx$, the expected value of the the standard normal multiplied by x .

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Integration by Parts

To find an integral of the form $\int_a^b f(x) \cdot g'(x) dx$ one can apply the integration by parts rule:

$$\int f(x) \cdot g'(x) dx = f(x) \cdot g(x) - \int g(x) \cdot f'(x) dx$$

Integration by Parts: Example

Find $\int \ln(x) dx$.

Let $f(x) = \ln(x)$ and $g'(x) = 1$. Then $f'(x) = \frac{1}{x} dx$ and $g(x) = x$. So

$$\begin{aligned}\int f(x) \cdot g'(x) dx &= f(x) \cdot g(x) - \int g(x) \cdot f'(x) dx = \\ &= x \ln(x) - \int x \cdot \frac{1}{x} dx = x[\ln(x) - 1]\end{aligned}$$

Integration by Parts: Example 2

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Integration by Parts: Example 2

Find $\int xe^x dx$

- How can we define xe^x in terms of $f(x)$ and $g'(x)$?
- $f(x) = x$ and $g'(x) = e^x$.
- Then we can redefine the integral as
$$xe^x - \int e^x dx = e^x(x - 1) + C$$